

NeuroCore Functional Stability Index (NFSI)

1. Physical Layer

$$V(t) = V1(t) - V2(t)$$

EEG is measured as a differential signal. This layer is immutable.

2. Dynamic Layer

Local Dynamics:

$$\Phi(t) = \sum \log(1 + |\nabla V(i)|)$$

Transition Sensitivity:

$$\Delta(t) = |\Phi(t) - \Phi(t-1)|$$

Global Coherence:

$$K(t) = \sigma / \mu$$

3. Asymptotic Stability

$$L(t) \rightarrow \phi \approx 0.55$$

Represents convergence toward a stable functional regime.

4. NeuroCore Functional Stability Index

$$\text{NFSI}(t) = w1 \cdot \Delta(t) + w2 \cdot K(t) + w3 \cdot |L(t) - \phi|$$

5. Interpretation

Low NFSI → Stable regime

Medium NFSI → Transition zone

High NFSI → Instability

6. Clinical Meaning (Anesthesia)

$\Delta \uparrow$ → transition (induction/emergence)

$K \uparrow$ → global instability

$|L - \phi| \uparrow$ → deviation from equilibrium

Conclusion

NeuroCore does not replace EEG measurement but defines a derived state-space representation capturing functional stability and transitions.